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CLOTHING PROCESSING APPARATUS

Abstract

A clothing processing apparatus includes a drum including a cylindrical-shaped drum body, front cover a rear cover, and a drum inlet through the front cover, a stationary panel spaced apart from the rear cover, a power transmission including a housing fixed to the stationary panel, an input shaft, an output shaft through the stationary panel, and a gear, and a motor including a stator and a rotor. The drum body, the rear cover, and the front cover are made of the same material, and the thickness of the drum body is configured to be smaller than the thickness of the rear cover and the thickness of the front cover.

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Background/Summary

TECHNICAL FIELD

[0001] This application relates to clothing processing apparatus.

BACKGROUND ART

[0002] A clothing processing apparatus is a generic term referring to a washing machine for washing clothing (an object to be washed or an object to be dried), a dryer for drying clothing, and an apparatus capable of performing both washing and drying of clothing.

[0003] A washing machine generally includes a tub configured to store water, a washing drum provided inside the tub to store clothing, and a driver (or washing driver) configured to rotate the washing drum, and a dryer generally includes a drying drum configured to store clothing, a driver (or drying driver) configured to rotate the drying drum, and a heat exchanger configured to supply air to the drying drum to remove moisture from the clothing.

[0004] The washing driver generally includes a stator fixed to a tub to form a rotating magnetic field, a rotor rotated by the rotating magnetic field, and a rotating shaft arranged through the tub to connect the washing drum and the rotor, while the drying driver generally includes a motor, a pulley fixed to a rotating shaft of the motor, and a belt (power transmitter) connecting the rotational motion of the pulley to the drying drum.

[0005] The washing driver is arranged such that the rotating shaft of the motor connects the washing drum and the rotor. In order to wash or spin clothes, the washing driver needs to control the rotational speed of the washing drum to be high or to change the rotation direction of the washing drum. By arranging the rotating shaft of the motor to directly connect the washing drum and the rotor, the rotational speed and rotation direction of the washing drum may be easily controlled.

[0006] The conventional drying driver generally has a structure in which a power transmitter such as a belt connects the drying drum and the rotating shaft of the motor. This is because the dryer rarely needs to maintain a high rotational speed of the drying drum or change the direction of rotation of the drying drum, and thus it is allowed to rotate the drying drum through a power transmitter such as a belt. However, if the speed and direction of rotation of the drying drum can be changed, the movement of the clothes inside the drying drum may be controlled, and thus the dryer can be expected to shorten the drying time and improve the drying performance.

[0007] A conventional dryer is equipped with a drying driver that connects the rotor and the drying drum with a reducer (power transmitter) (Korean Patent Application Publication No. 10-2020-0065931). The drying driver is configured by coaxially arranging an input shaft connected to the rotor and an output shaft connected to the drying drum, such that the drying time may be shortened or the drying performance may be improved.

[0008] The above-described dryer can control the number of rotations of the drying drum in various ways as well as change the direction of rotation. Accordingly, the durability of the drying drum to which the rotating shaft is fixed and the durability of the panel supporting the drying drum are likely to be reduced.

[0009] Furthermore, since the dryer configured as described above allows the vibration of the drum to be transmitted to the reducer and a fixing panel to which the reducer is fixed, the durability of the reducer and the durability of the fixing panel may be reduced due to vibration of the drum.

[0010] In addition, lubricating oil is stored inside the reducer provided in the above-described dryer. In the case of a structure in which the drying drum rotates around a rotating shaft arranged

horizontal to the ground, a mating surface of the housing is located on the circumferential surface of the reducer housing, and there is a possibility that lubricating oil leaks to the outside through the mating surface.

DISCLOSURE

Technical Problem

[0011] An object of the present application is to provide a clothing processing apparatus capable of minimizing the deterioration of the durability of a reducer and the deterioration of the durability of a fixing panel on which the reducer is fixed.

[0012] Another object of the present application is to provide a clothing processing apparatus capable of minimizing the risk of leakage of lubricating oil from a reducer having an output shaft arranged parallel to the ground or an output shaft inclined at an angle of less than **90** degrees with respect to the ground.

[0013] Another object of the present application is to provide a clothing processing apparatus capable of reducing the rotational speed of a rotor and transmitting the reduced rotational speed to a drum with the center of rotation of the rotor and the center of rotation of the drum kept concentrically aligned.

Technical Solution

[0014] In accordance with one aspect of the present disclosure, provided is a clothing processing apparatus including a drum including a drum body having a cylindrical shape with two opposing sides open, a front cover arranged to close one of the open sides of the drum body to define a front surface of the drum body, a rear cover arranged to close the other one of the open sides of the drum body to define a rear surface of the drum body, and a drum inlet formed through the front cover in a penetrating manner to communicate with an interior of the drum body, a fixing panel arranged spaced apart from the rear cover, a power transmitter including a housing fixed to the fixing panel, an input shaft having one end positioned outside the housing and an opposite end positioned inside the housing, an output shaft arranged through the fixing panel, the output shaft having one end fixed to the rear cover and an opposite end positioned inside the housing, and a gear unit disposed inside the housing to transmit rotational motion of the input shaft to the output shaft, and a motor including a stator fixed to the fixing panel or the housing to form a rotating magnetic field, and a rotor rotated by the rotor, the one end of the input shaft being fixed to the rotor.

[0015] The drum body, the rear cover, and the front cover are formed of a same material, and a thickness of the drum body may be set to be less than a thickness of the rear cover and a thickness of the front cover.

[0016] A thickness of the fixing panel may be set to be greater than the thickness of the rear cover.

[0017] The clothing processing apparatus may further include a ring-shaped panel bent portion formed by bending the fixing panel in a direction toward the rear cover or in a direction away from the rear cover to surround the output shaft.

[0018] The panel bent portion may be formed by bending the fixing panel in the direction toward the fixing panel, and the housing may be positioned inside the panel bent portion.

[0019] The clothing processing apparatus may further include a drum bent portion formed by bending the rear cover in a direction toward the front cover or in a direction toward the fixing panel, the output shaft being fixed to the drum bent portion.

[0020] Among the thickness of the drum body, the thickness of the rear cover, and the thickness of the front cover, the thickness of the front cover may be set to be the greatest.

[0021] Among the thickness of the drum body, the thickness of the rear cover, and the thickness of the front cover, the thickness of the rear cover may be set to be the greatest.

[0022] The clothing processing apparatus may further include a drum support rotatably supporting a circumferential surface of the front cover or a circumferential surface of the drum body.

[0023] The input shaft and the output shaft may form a concentric axis. The input shaft and the output shaft may be parallel to the ground.

[0024] The input shaft and the output shaft may form a concentric axis. The input shaft and the output shaft may be inclined at an angle of less than 90 degrees with respect to a ground.

[0025] The housing may include a housing base provided with an input shaft through-hole allowing the input shaft to be arranged therethrough in a penetrating manner, an input shaft support provided to the housing base to surround the input shaft through-hole, a housing circumferential surface extending from an edge of the housing base toward the fixing panel to define an accommodation space, the gear unit being mounted in the accommodation space, a cover body fixed to the housing circumferential surface to close the accommodation space, an output shaft through-hole formed through the cover body in a penetrated manner, the output shaft being inserted into the output shaft through-hole, an output shaft support protruding from the cover body toward the fixing panel to surround the output shaft through-hole, and an oil leakage preventer configured to prevent lubricating oil supplied to the gear unit from being discharged from the accommodation space.

[0026] The oil leakage preventer may include a first sealing portion disposed to seal a space between the input shaft support and the input shaft, a second sealing portion disposed to seal a space between the output shaft support and the output shaft, and a third sealing portion disposed to seal a space between the housing circumferential surface and the cover body.

[0027] The clothing processing apparatus may further include a cover fixing plate provided on the housing circumferential surface, the cover fixing plate extending away from the housing circumferential surface in a radial direction of the input shaft through-hole such that the cover body is fixed to the cover fixing plate. The third sealing portion may include a ring-shaped sealing protrusion protruding from the cover body toward the cover fixing plate, and a ring-shaped accommodation groove provided in the cover fixing plate, the sealing protrusion being accommodated in the accommodation groove.

Advantageous Effects

[0028] The present application provides a clothing processing apparatus capable of minimizing the deterioration of the durability of a reducer and the deterioration of the durability of a fixing panel on which the reducer is fixed.

[0029] Furthermore, the present application provides a clothing processing apparatus capable of minimizing the risk of leakage of lubricating oil from a reducer having an output shaft arranged parallel to the ground or an output shaft inclined at an angle of less than 90 degrees with respect to the ground.

[0030] Furthermore, the present application provides a clothing processing apparatus capable of reducing the rotational speed of a rotor and transmitting the reduced rotational speed to a drum with the center of rotation of the rotor and the center of rotation of the drum kept concentrically aligned.

Description

DESCRIPTION OF DRAWINGS

[0031] FIG. 1 shows an example of a clothing processing apparatus.

[0032] FIG. 2 shows an example of a cabinet structure and a support panel.

[0033] FIG. 3 shows an example of a drum.

[0034] FIG. 4 shows an example of a dryer.

[0035] FIG. 5 shows an example of a fixing panel.

[0036] FIGS. 6 and 7 show an example of a power transmitter.

[0037] FIG. 8 shows an example of a coupling structure of the power transmitter, the fixing panel, and the drum.

[0038] FIGS. 9 and 10 show the structure of the drum.

[0039] FIG. 11 shows an example of an oil leakage preventer.

BEST MODE

[0040] Hereinafter, embodiments of a clothing processing apparatus will be described in detail with reference to the accompanying drawings.

[0041] FIG. 1 shows an example of a clothing processing apparatus **100**. The clothing processing apparatus **100** may include a cabinet **1**, and a drum **2** rotatably disposed in the cabinet to provide a space in which clothing (to be washed or dried) is stored. As shown in FIG. 2, a dryer **3** may be provided in the cabinet **1** to supply hot dry air (air having a temperature higher than room temperature, air having a higher degree of dryness than the dryness of room air) to the drum **2** to remove moisture from the clothing.

[0042] The cabinet **1** includes a front panel **11** forming a front surface of the clothing processing apparatus, and a base panel **17** forming a bottom surface of the clothing processing apparatus. The front panel **11** is provided with an inlet **111** communicating with the drum **2**, the inlet **111** may be arranged to be closed by a door **113**.

[0043] The front panel **11** is provided with a control panel **115**. The control panel **115** may include an input unit configured to receive a control command from a user and a display configured to output information such as a control command selectable by the user. The input unit may include a power supply requester configured to make a request for power supply to the clothing processing apparatus, a course input unit configured to enable the user to select a desired course among a plurality of courses, and an execution requester configured to request initiation of the course selected by the user.

[0044] The drum **2** may have a hollow cylindrical shape. As shown in FIG. 3, the drum **2** may include a cylindrical drum body **21** having an open front side and an open rear side, a front cover **22** (defining a front surface of the drum) fixed to the drum body **21** to close the front side, and a rear cover **23** (defining a rear surface of the drum) fixed to the drum body **21** to close the rear side.

[0045] The front cover **22** may be provided with a drum inlet **221** allowing the inside of the drum body **21** to communicate with the outside therethrough, and the rear cover **23** may be provided with an air inlet **233** allowing external air to be introduced into the drum body **21** therethrough.

[0046] The drum body **21** may further include a lifter **24**. The lifter **24** is a means for causing the clothing to repeatedly rise and fall in the drum. The lifter **24** may be provided as a plate extending from the front cover **22** toward the rear cover **23** protrudes from the drum body **21** toward the center of rotation of the drum **2** (i.e., from a circumferential surface of the drum toward the center of rotation of the drum).

[0047] When the clothing processing apparatus **100** is configured as an apparatus dedicated to drying clothing, the drum **2** is not provided with a drum through-hole formed through the drum body **21** to allow the inside of the drum to communicate with the outside of the drum.

[0048] When the clothing processing apparatus **100** is provided as an apparatus to dry and wash clothing, a tub to store water may be provided in the cabinet, and the drum may be rotatably arranged in the tub. In this case, the drum body may need to have a drum through-hole that allows the tub and the drum to communicate with each other.

[0049] As shown in FIG. 4, the drum **2** described above may be rotatably fixed to at least one of a first body support **12** and a second body support **15**. FIG. 4 illustrates an example in which the rear cover **23** is rotatably fixed to the second body support **15** by a power transmitter **6**, which will be described later, and the front cover **22** is rotatably connected to the first body support **12**.

[0050] The first body support **12** may include a support panel **121** fixed to the cabinet **1** and positioned between the front panel **11** and the front cover **22**. The support panel **121** may be fixed to the base panel **17** so as to be positioned between the front panel **11** and the front cover **22**.

[0051] The support panel **121** may include a support panel through-hole **122**, a drum connection body **123** connecting the support panel through-hole **122** to the drum inlet **221**, and a panel connection body **125** connecting the support panel through-hole **122** and the inlet **111**. The support panel through-hole **122** is formed through the support panel **121** to allow the inlet **111** and the drum inlet **221** to communicate with each other.

[0052] The drum connection body **123** may be provided as a pipe fixed to the rear surface of the support panel **121** (i.e., the surface facing the drum inlet in the space provided by the support panel). One end of the drum connection body **123** may be arranged to surround the support panel through-hole **122**, and a free end of the drum connection body **123** may be arranged to support the front cover **22**. That is, the free end of the drum connection body **123** may be inserted into the drum inlet **221**, or may be arranged to be in contact with a free end (edge of the drum inlet) of the front cover **22** defining the drum inlet **221**. FIG. **4** illustrates an example in which the free end of the drum connection body **123** contacts the free end of the front cover **22**.

[0053] The panel connection body **125** may be provided as a pipe fixed to the front surface of the support panel **121** (i.e., the surface facing the front panel in the space provided by the support panel). One end of the panel connection body **125** may be arranged to surround the support panel through-hole **122**, and the opposite end of the panel connection body **125** may be connected to the inlet **111**. Accordingly, the clothing supplied through the inlet **111** may move to the drum body **21** through the panel connection body **125**, the support panel through-hole **122**, the drum connection body **123**, and the drum inlet **221**.

[0054] An exhaust port **126** for discharging the air from the drum **2** to the outside of the drum may be provided in the support panel **121**, and a filter **127** may be detachably fixed to the exhaust port **126**. The filter **127** may be formed in any structure capable of filtering foreign substances from the air moving from the drum **2** to the exhaust port **126**.

[0055] The support panel **121** may further include a drum support **128** to prevent the drum **2** from sagging. The drum support may include a roller fixed to the support panel **121** to rotatably support the drum **2**. The drum support **128** may support a region of the drum located below a horizontal line passing through the center of the drum inlet.

[0056] Although FIG. **4** illustrates a case where the drum support **128** is arranged to support the front cover **22**, the drum support **128** may be arranged to support the circumferential surface of the drum body **21**.

[0057] The second body support **15** may include a fixing panel **151** fixed to the cabinet **1** to be positioned at a point spaced apart from the rear cover **23**. FIG. **5** illustrates an example in which the fixing panel **151** is fixed to the base panel **17** to define the rear surface of the clothing processing apparatus **100** (i.e., the rear surface of the cabinet).

[0058] The fixing panel **151** may be provided with a panel bent portion **152**, which provides a space in which the motor **5** is mounted. The panel bent portion **152** may be provided as a recess formed by concavely bending the fixing panel **151** to a side on which the rear cover **23** of the drum is positioned. The fixing panel **151** may be provided with a fixing panel through-hole **153** through which a shaft (i.e., an output shaft) for rotating the drum **2** is arranged. The fixing panel through-hole **153** may be positioned inside the panel bent portion **152**.

[0059] As described above, as the drum **2** includes the drum body **21**, the front cover **22** fixed to the drum body, and the rear cover **23** fixed to the drum body, the rigidity of the drum may be increased compared to a structure in which the open front side and rear side of the drum body **21** are rotatably connected to the support panel **121** and fixing panel **151**, respectively. By increasing the rigidity of the drum, the deformation of the drum body **21** may be minimized when the drum is rotated, which may suppress the issue of clothing being stuck in the space between the drum body and the support panel and the space between the drum body and the fixing panel (and may minimize the load on the motor) when the drum body **21** is deformed.

[0060] As shown in FIG. **4**, the dryer **3** may include an exhaust passage **31** connected to the exhaust outlet **126**, a supply passage **32** guiding air supplied from the exhaust passage **31** to the drum body **21**, and a heat exchanger **34** arranged in the exhaust passage **31** to sequentially perform dehumidification and heating of air.

[0061] The exhaust passage **31** may include a first duct **311** connected to the exhaust port **126**, a second duct **312** connected to the supply passage **32**, and a third duct **313** connecting the first duct

311 and the second duct **312**. The third duct **313** may be fixed to the base panel **17**.

[0062] Various devices capable of sequentially performing dehumidification and heating of the air introduced into the exhaust passage **31** may be provided as the heat exchanger **34**. FIG. **4** shows an example in which a heat pump is provided as the heat exchanger **34**. That is, the heat exchanger **34** includes a first heat exchanger (heat absorber) **341** configured to remove moisture from the air entering the exhaust passage **31**, a second heat exchanger (heater) **343** arranged in the exhaust passage **31** to heat the air passed through the heat absorber, **341**, and a fan **349** configured to cause the air discharged from the drum **2** to sequentially pass through the heat absorber and heater and then to move to the supply duct **32**.

[0063] The heat absorber **341** and the heater **343** are sequentially disposed along the direction of movement of air and are connected to each other by a refrigerant pipe **348** defining a circulation passage for refrigerant. The refrigerant is moved along the refrigerant pipe **348** by a compressor **345** positioned outside the exhaust passage **31**, and the refrigerant pipe **348** is provided with a pressure regulator **347** to regulate the pressure of the refrigerant.

[0064] As shown in FIG. **5**, the air inlet **233** provided in the rear cover **23** of the drum may be provided with a plurality of holes disposed to surround the center of the rear cover **23** (the center of rotation of the drum). In this case, the supply passage **32** may include a supply duct **321** provided in the fixing panel **151** to form a path for the movement of air discharged from the second duct **312**, a first flow passage definer **323** and a second flow passage definer **324** to guide the air inside the supply duct **321** to the air inlet **233** by blocking the air inside the supply duct **321** from leaking to the outside.

[0065] The supply duct **321** may be configured to form a flow passage (a movement path of air) by bending the fixing panel **151** in a direction away from the rear cover **23**. Further, the supply duct **321** may be formed in a ring shape to surround the panel bent portion **152**, and an outlet of the second duct **312** may be connected to a circumferential surface of the supply duct **321**.

[0066] The first flow passage definer **323** may be arranged to surround an outer circumferential surface of the ring formed by the air inlets **233** (a circumferential surface having a longer diameter of the two circumferential surfaces of the ring), and the second flow passage definer **324** may be arranged to surround an inner circumferential surface of the ring formed by the air inlets **233** (a circumferential surface having a shorter diameter of the two circumferential surfaces of the ring).

[0067] The first and second flow passage definers **323** and **324** may be fixed to the rear cover **23**, or they may be fixed to the supply duct **321**. FIG. **5** illustrates an example case where the flow passage definers **323** and **324** are fixed to the rear cover **23**.

[0068] In the case of FIG. **5**, the free end of the first flow passage definer **323** is arranged to surround an outer circumferential surface of the flow passage (ring-shaped flow passage) defined by the supply duct **321**, and the free end of the second flow passage definer **324** is arranged to surround an inner circumferential surface of the flow passage defined by the supply duct **321**. The first flow passage definer **323** and second flow passage definer **324** may be formed of rubber, felt or the like.

[0069] The motor **5**, which is configured to generate power required for rotation of the drum **2**, includes a stator **51** positioned at the panel bent portion **152** to form a rotating field, and a rotor **52** rotated by the rotating magnetic field. The rotational motion of the rotor **52** is transmitted to the drum **2** via the power transmitter **6** fixed to the fixing panel **151**, and the stator **51** is fixed to at least one of the fixed panel **151** or the power transmitter **6**.

[0070] In order to prevent the motor **5** positioned in the panel bent portion **152** from being exposed to the outside (in order to improve the durability of the motor and prevent safety accidents by preventing the motor from being exposed to the outside environment), the fixing panel **151** may further include a cover panel **19** arranged to prevent the motor **5** from being exposed to the outside. Further, the cover panel **19** may be formed in a shape capable of preventing the supply duct **321** from being exposed to the outside (a shape surrounding the supply duct). This is intended to

minimize the dissipation of heat to the outside of the supply duct **321**, as well as to prevent accidents that may occur when a human body comes into contact with the supply duct **321**.

[0071] The stator **51** includes a ring-shaped core **511** having a through-hole (core through-hole) at a center thereof, a plurality of support bars **513** radially protruding from an outer circumferential surface of the core **511**, an insulator **512** insulating the core and support bars, and a coil **514** arranged to surround the support bars **513**.

[0072] The rotor **52** may include a disk-shaped rotor body **521**, a rotor circumferential surface **522** extending from an edge of the rotor body **521** toward the fixing panel **515** to form a space in which the stator **51** is accommodated, and a plurality of permanent magnets **523** fixed to the rotor circumferential surface **522** such that N and S poles are alternately exposed.

[0073] FIG. **6** shows an example of the power transmitter **6** connecting the rotor **52** to the rear cover **23** of the drum.

[0074] The power transmitter **6** includes a housing **H** fixed to the fixing panel **151**, an input shaft **64** rotatably fixed to a bottom surface (i.e., the surface facing the position where the rotor is positioned) of the housing **H**, an output shaft **65** rotatably fixed to a top surface (i.e., the surface facing the position where the fixing panel is positioned) of the housing **H**, and a gear unit **G** disposed in the housing to transmit rotational motion of the input shaft **64** to the output shaft **65**.

[0075] The input shaft **64** may be provided as a shaft having one end fixed to the rotor **52** and an opposite end positioned inside the housing **H**. The output shaft **65** may be provided as a shaft having one end fixed to the rear cover **23** and an opposite end positioned inside the housing **H**. The gear unit **G** is arranged to connect one end of the input shaft **64** positioned inside the housing **H** and one end of the output shaft **65** positioned inside the housing **H**.

[0076] The housing **H** is fixed to the fixing panel **151** so as to be positioned in a space (space outside the cabinet) separated from the space (space inside the cabinet) in which the drum **2** is positioned. This is intended to minimize the transfer of heat inside the cabinet (heat emitted by the drum or the dryer) to the inside of the housing **H** and thus improve the durability of the power transmitter **6**.

[0077] The input shaft **64** may be coupled to the rotor body **521** by a shaft fixing portion **54**. The shaft fixing portion **54** may include an shaft fixing hole **541** to which one end of the input shaft **64** is fixed.

[0078] The output shaft **65** may be inserted into the fixing panel through-hole **153** and connected to the rear cover **23**, which may be provided with a shaft fastening portion **25**. The output shaft **65** is fixed to the shaft fastening portion. This is intended to decentralize the stress acting on the center of the rear cover **23** when the output shaft **65** is rotated.

[0079] The shaft fastening portion **25** may be provided in a drum bent portion **235**, which is provided in the rear cover **23**. The drum bent portion **235** may be provided as a groove formed by bending the rear cover **23** toward the front cover **22** to form a ring surrounding the output shaft **65**. The drum bent portion **235** may attenuate transmission of the vibration of the rear cover **23** to the output shaft **65**.

[0080] Since the panel bent portion **152** provided in the fixing panel **151** is capable of minimizing the transmission of the vibration of the housing **H** to other areas of the fixing panel **151**, the clothing processing apparatus **100** provided with the drum bent portion **235** or panel bent portion **152** may minimize transmission of vibration of the drum to the power transmitter **6** (which may improve the durability of the drum, the fixing panel, and the power transmitter).

[0081] In order to minimize deformation of the fixing panel **151**, the housing **H** may be fixed to the fixing panel **151** via a transmitter bracket **61** and the housing fastening portion.

[0082] As shown in FIG. **5**, the transmitter bracket **61** may be provided with a bracket through-hole **611** through which the output shaft **65** is arranged, and the housing fastening portion may be provided with a bolt connecting the housing **H** and the transmitter bracket **61**. The transmitter bracket **61** may be formed of the same material as the fixing panel **151**, or may be formed of a

material having a strength higher than that of the fixing panel **151**.

[0083] As shown in FIG. **6**, the housing **H** may include a housing body **62** having a hollow cylindrical shape and having an open hole in a surface facing the fixed body **151**, and a housing cover **63** fixed to the housing body **62** to close the open hole.

[0084] The housing body **62** may be formed to define an accommodation space **621** in which the gear unit **G** is mounted. The accommodation space **621** may be defined by a housing base **62a** to which the input shaft **64** is fixed, and a housing circumferential surface **62b** extending from an edge of the housing base **62a** toward the housing cover **63**.

[0085] As shown in FIG. **7**, the housing body **62** may be provided with an input shaft support **625** extending from the housing base **62a** toward the rotor **52**. The input shaft support **625** may be provided as a pipe surrounding an input shaft through-hole **626**, which is formed through the housing body **62**.

[0086] The input shaft **64** inserted into the input shaft through-hole **626** is rotatably fixed to the input shaft support **625** through input shaft bearings **628** and **629**. The input shaft bearings may include a first input shaft bearing **628** fixed to the input shaft support **625**, and a second input shaft bearing **629** fixed inside the input shaft through-hole **626** and positioned between the first input shaft bearing **628** and the rotor **52**.

[0087] The housing cover **63** may be formed in any shape as long as it is capable of opening and closing an opening provided in the housing body **62**. FIG. **7** illustrates an example case where the housing cover **63** is provided with a disk-shaped cover body **631**. The housing cover **63** may be fixed to the housing body **62** through a cover fixing plate **623** provided on the housing circumferential surface **62b**.

[0088] The housing cover **63** may include an output shaft through-hole **632** formed through the cover body **631** in a penetrating manner, and a pipe-shaped output shaft support **635** extending from the cover body **631** toward the fixed panel **151** to surround the output shaft through-hole **632**.

[0089] The output shaft support **635** may be provided with output shaft bearings **638** and **639** rotatably fixing the output shaft **65** in the output shaft through-hole **632**.

[0090] The output shaft bearings may include a first output shaft bearing **638** and a second output shaft bearing **639** that are fixed to the output shaft support **635** and are positioned inside the output shaft through-hole **631**.

[0091] An edge of the cover body **631** may be provided with a mounting portion **637** to which the stator **51** is fixed. In this case, the stator **51** may be fixed to the housing **H** by bolts coupling the insulator **512** to the mounting portion **637**.

[0092] The gear unit **G** may include a ring gear **66** fixed to the housing circumferential surface **62b** and positioned in the accommodation space **621**, a drive gear **69** fixed to the input shaft **64** and positioned in the accommodation space **621**, a cage **67** positioned in the accommodation space **622** and having an opposite end of the output shaft **65** fixed thereto, and a driven gear **68** rotatably fixed to the cage **67** to connect the drive gear **69** and the ring gear **66**.

[0093] The ring gear **66** may include a gear body **661** fixed to the housing circumferential surface **62b**, a gear body through-hole **663** formed through the gear body in a penetrating manner, and gear teeth **665** provided along an inner circumferential surface of the gear body.

[0094] The cage **67** may include a first base **671** positioned inside the gear body through-hole **663**, one end of the output shaft **65** fixed to the first base, a second base **672** positioned inside the gear body through-hole **663** and having a base through-hole **673** at the center thereof, and a connection shaft **675** connecting the first base and the second base and forming an axis of rotation of the driven gear **68**. Since the output shaft **65** is fixed to the first base **671**, rotation of the output shaft **65** is determined depending on whether the cage **67** is rotated.

[0095] The driven gear **68** may include a plurality of gears rotatably fixed to the connection shaft **675**. The figure illustrates a case where the driven gear includes a first driven gear **681**, a second driven gear **682**, and a third driven gear **683**.

[0096] The input shaft **64** is inserted into the base through-hole **673** to form a concentric axis with the output shaft **65**, and the gear teeth provided to the drive gear **69** are positioned in a space formed between the driven gears to be engaged with the gear teeth of the driven gears **681**, **682**, and **683**.

[0097] In order to decentralize the stress acting on the center of the rear cover **23** when the output shaft **65** rotates (i.e., in order to minimize the deterioration of the durability of the rear cover), the output shaft **65** is connected to the rear cover **23** of the drum through the shaft fastening portion **25**.

[0098] As shown in FIG. **8**, the rear cover **23** may be provided with a rear cover through-hole **231**, and the shaft fastening portion **25** may be positioned at the drum bent portion **235** of the rear cover **23** to close the rear cover through-hole **231**.

[0099] The shaft fastening portion **25** is provided with a fastening portion through-hole **251** into which the output shaft **65** is inserted. The output shaft **65** inserted into the fastening portion through-hole **251** may be fixed to the shaft fastening portion **25** by a nut **252**. To this end, a screw thread **651** should be provided on the circumferential surface of the output shaft.

[0100] In order to prevent the shaft fastening portion **25** from being exposed to the inside of the drum **2**, the clothing processing apparatus **100** may further include a fastening portion cover **27**. The fastening portion cover **27** may be formed in any shape capable of closing the rear cover through-hole **231** and preventing exposure of the shaft fastening portion **25**.

[0101] The fastening portion cover **27** may be fixed to the output shaft **65** by a bolt **273**. To this end, the fastening portion cover **27** may be provided with a cover through-hole **271** into which the bolt **273** is inserted, and a fastening groove **652** to which the bolt is fastened may be provided at one end of the output shaft **65**.

[0102] In order to facilitate the coupling between the output shaft **65** and the shaft fastening portion **25**, a washer **653** may be further arranged on the output shaft **65**. The washer **653** has a diameter larger than the diameter of the free end of the output shaft **65** (i.e., one end of the output shaft on which the screw thread is formed), and the center thereof is provided with a washer through-hole **654** into which the free end of the output shaft **65** is inserted.

[0103] The washer **653** may allow the rear cover **23** of the drum and the output shaft **65** to be connected to each other at a right angle (which improves assemblability), and reduce the risk of separation of the nut **252** from the screw thread **651** due to vibration.

[0104] The operation of the power transmitter **6** is described below. When the rotor **52** rotates, the input shaft **64** rotates. When the drive gear **69** is rotated by the input shaft **64**, the driven gears **681**, **682**, and **683** engaged with the drive gear **69** also rotate. Since the driven gears **681**, **682**, and **683** are engaged with the ring gear **66** fixed to the housing body **62**, the cage **67** and the output shaft **65** and the drum **2** fixed to the output shaft **65** will rotate when the driven gear **68** rotates.

[0105] When the drum **2** rotates in the clothing processing apparatus **100** having the above-described structure, vibration is transmitted to the power transmitter **6** and the fixing panel **151**. When the drum rotates, the clothing is repeatedly lifted and dropped inside the drum, and the vibration generated in the drum is transmitted to the gear unit **G** and the housing **H** through the output shaft **65**. Then, the vibration of the housing **H** will be transmitted to the fixing panel **151**.

[0106] In order to minimize the deterioration of the durability of the power transmitter **6** including the gear unit **G** and the durability of the fixing panel **151**, the drum **2** may have a structure capable of absorbing some of vibrations generated when the clothing falls inside the drum.

[0107] As described above, the drum **2** is formed by coupling the drum body **21**, the front cover **22**, and the rear cover **23**. In this case, most of the clothing dropped by the rotation of the drum may collide with the drum body **21**. Therefore, by setting the rigidity of the drum body **21** to be lower than the rigidity of the front cover **22** and the rigidity of the rear cover **23**, the drum body **21** may absorb some of the force generated when the clothing collides with the drum body.

[0108] FIG. **9** shows an example of reducing the rigidity of the drum body **21** when the drum body **21**, the front cover **22**, and the rear cover **23** are formed of the same material.

[0109] In this case, the thickness **t1** of the drum body **21** may be set to be less than the thickness **t3** of the rear cover **23** and the thickness **t2** of the front cover **22**. When the thickness **t1** of the drum body **21** is set to be less than the thicknesses **t2** and **t3** of the two covers, some of the force input to the drum **2** when the clothing is dropped may be absorbed by the drum body **21**. Accordingly, the drum **2** having the structure of FIG. **9** may minimize the transmission of external force to the housing **H** and the fixing panel **151**.

[0110] Since the front cover **22** is provided with the drum inlet **221**, the rigidity of the front cover **22** may be lower than that of the drum body **21** or the rear cover **23**. To compensate for this issue, the thickness **t3** of the front cover **22** may be set to be the greatest in the drum **2**. This setting may be applied even in the case where the front cover **22** is supported by the drum support **128** described above.

[0111] However, in the case where the circumferential surface of the front cover **22** is supported by the drum support **128**, the thickness **t3** of the rear cover **23** may be set to be the greatest in the drum **2** because the front cover **22** is less likely to be deformed. This is because the rear cover **23** needs to be configured to transmit the force supplied from the output shaft **65** to the drum body **21**. When the thickness **t3** of the rear cover is set to be the greatest, the transmission of the vibration of the drum body **21** to the output shaft **65** may be expected to be minimized. In more detail, referring to FIG. **10**, vibration generated inside the drum **2** may be first

[0112] reduced by the drum body **21** having the least thickness. The vibration not absorbed by the drum body **21** will be absorbed by the rear cover **23**, which is thicker than the drum body **21**, and the vibration not absorbed by the rear cover **23** will be transmitted to the housing **H** of the power transmitter **6** via the output shaft **65**.

[0113] The housing **H** is fixed to the fixing panel **151**, and a portion of the vibration transmitted to the housing **H** is absorbed by the fixing panel **151**. Accordingly, the clothing processing apparatus having the above-described structure may minimize the transmission of the vibration from the inside the drum **2** to the gear unit **G** provided inside the housing **H**.

[0114] Since the housing **H** of the power transmitter **6** is fixed to the panel bent portion **152**, the clothing processing apparatus having the above-described structure may also minimize the transmission of the vibration transmitted to the housing **H** through the output shaft **65** to the other carriers of the fixed panel **15**.

[0115] Therefore, the clothing processing apparatus **100** may minimize the deterioration of the durability of the power transmitter **6** and the deterioration of the durability of the fixing panel **151** due to the vibration generated during the rotation of the drum.

[0116] The output shaft **65** and the input shaft **64** included in the power transmitter **6** form a concentric axis. The output shaft **65** and the input shaft **64** are arranged parallel to the ground (the bottom surface of a space in which the cabinet is installed) (as indicated by **A1**). This means that the contact surfaces of the housing body **62** and the housing cover **63** are perpendicular to the ground. Due to such structural features, the lubricating oil supplied to the gear unit **G** may be leaked through the contact surfaces of the housing body **62** and the housing cover **63**.

[0117] In order to prevent the lubricating oil inside the accommodation space **621** from leaking to the outside of the housing **H**, the power transmitter **6** may further include an oil leakage preventer **7**.

[0118] FIG. **11** shows an embodiment of the oil leakage preventer **7**. The oil leakage preventer **7** may include a first sealing portion **71** for sealing a space between the input shaft support **625** and the input shaft **64**, a second sealing portion **72** for sealing a space between the output shaft support **635** and the output shaft **65**, and a third sealing portion **73** for sealing the contact surfaces of the housing circumferential surface **62b** and the cover body **631**.

[0119] The first sealing portion **71** may be a ring-shaped gasket fixed inside the input shaft support **625**. In this case, the first sealing portion **71** may be disposed between the first input shaft bearing **628** and the second input shaft bearing **629**, or may be disposed between the second input shaft

bearing **629** and the free end **F1** of the input shaft support **625**.

[0120] The second sealing portion **72** may be a ring-shaped gasket fixed in the output shaft support **635**. In this case, the second sealing portion **72** may be disposed between the first output shaft bearing **638** and the second output shaft bearing **639**, or may be disposed between the second output shaft bearing **639** and the free end **F2** of the output shaft support **635**.

[0121] Unlike the configuration illustrated in the figure, when the input shafts bearing and the output shaft bearing are each provided as a single bearing, the first sealing portion **71** may be disposed between the input shaft bearing and the free end **F1** of the input shaft support, and the second sealing portion **72** is disposed between the output shaft bearing and the free end **F2** of the output shaft support.

[0122] The third sealing portion **73** may be configured in various structures capable of sealing the circumferential surface **62b** of the housing and the contact surface of the cover body **631**. In the housing **H** of FIG. **11**, the cover body **631** is connected to the housing body **62** by the cover fixing plate **623**. Therefore, the third sealing portion **73** may seal the edge of the cover fixing plate **623** and the cover body **631** (i.e., a region contacting the cover fixing plate).

[0123] As shown in the figure, the third sealing portion **73** may be provided as a ring-shaped gasket **733** positioned between the cover body **631** and the cover fixing plate **623**.

[0124] Alternatively, the third sealing portion **73** may include a ring-shaped sealing protrusion **731** protruding from one of the cover body **631** and the cover fixing plate **623** toward the other, and a sealing groove **732** provided on the other of the cover body **631** and the cover fixing plate **623** to accommodate the sealing protrusion **731**. In the example shown in the figure, the cover body **631** is provided with the sealing protrusion **731**, and the cover fixing plate **623** is provided with the sealing groove **732**. The sealing protrusion **731** and the sealing groove **732** may be bonded to each other through thermal bonding.

[0125] Although not shown in the figure, the number of sealing protrusions **731** and the number of sealing grooves **732** may each be two or more. That is, the sealing protrusion **731** includes a first protrusion having a ring shape around the output shaft through-hole **632** and a second protrusion having a ring shape with a diameter greater than a diameter of the first protrusion, and the sealing groove **732** may include a first groove in which the first protrusion is accommodated, and a second groove in which the second protrusion is accommodated.

[0126] In addition, the third sealing portion **73** may include the sealing protrusion **731**, the sealing groove **732**, and the gasket **733**. In this case, the gasket **733** may be arranged to surround the sealing protrusion **731**, or the sealing protrusion **731** may be arranged to surround the gasket **733**.

[0127] The first sealing portion **71** and the second sealing portion **72** may be formed of a material different from that of the housing **H**. In this case, since the thermal strain of the sealing portions **71** and **72** is different from that of the housing **H**, the sealing portions **71** and **72** may be separated from the housing **H** in some cases.

[0128] In order to prevent the first sealing portion **71** and the second sealing portion **72** from being separated from the housing **H**, the clothing processing apparatus **100** may further include sealer fixing portions **74** and **75**.

[0129] The sealer fixing portion may include a first fixing portion **74** configured to prevent the first sealing portion **71** from being separated from the input shaft support **625**, and a second fixing portion **75** configured to prevent the second sealing portion **72** from being separated from the output shaft support **635**.

[0130] The first fixing portion **74** may be provided as a protrusion or a ring-shaped rib protruding from the input shaft support **625** toward the center of the input shaft through-hole **626** (i.e., the circumferential surface of the input shaft). In this case, the distance **L1** from the circumferential surface of the input shaft support **625** to the free end of the first fixing portion **74** may be longer than the distance from the circumferential surface of the input shaft support **625** to the circumferential surface of the first sealing portion **71**, and may be shorter than the distance from the

circumferential surface of the input shaft support **625** to the circumferential surface of the input shaft **64**.

[0131] Similarly, the second fixing portion **75** may be provided as a protrusion or a ring-shaped rib protruding from the output shaft support **635** toward the center of the output shaft through-hole **632** (i.e., the circumferential surface of the output shaft). In this case, the distance **L2** from the circumferential surface of the output shaft support **635** to the free end of the second fixing portion **75** may be longer than the distance from the circumferential surface of the output shaft support **635** to the circumferential surface of the second sealing portion **72**, and may be shorter than the distance from the circumferential surface of the output shaft support **635** to the circumferential surface of the output shaft **65**.

[0132] Although the embodiments have been described based on the clothing processing apparatus **100** in which the drum **2** rotates about a shaft parallel to the ground, the structure of the drum **2** and the oil leakage preventer **7** may be applied even when the shaft of the drum is inclined with respect to the ground. That is, as shown in FIG. **9**, the drum **2** and the oil leakage preventer **7** may be applied even when a straight line connecting the input shaft and the output shaft is inclined at an angle **A2** of less than **90** degrees with respect to the ground.

[0133] It should be noted that the above-described clothing processing apparatus can be modified in various forms, and therefore the scope of the present disclosure is not limited to the above-described embodiments.

Claims

1. A clothing processing apparatus comprising: a drum comprising: a drum body having a cylindrical shape with two opposing sides open; a front cover arranged to close one of the open sides of the drum body to define a front surface of the drum body; a rear cover arranged to close the other one of the open sides of the drum body to define a rear surface of the drum body; and a drum inlet formed through the front cover in a penetrating manner to communicate with an interior of the drum body; a fixing panel apart from the rear cover; a power transmitter comprising: a housing fixed to the fixing panel; an input shaft with one end located outside the housing and another end located inside the housing; an output shaft with one end formed to pass through the fixing panel and fixed to the rear cover and another end positioned inside the housing; and a gear unit disposed inside the housing to transmit rotational motion of the input shaft to the output shaft; and a motor comprising: a stator fixed to the fixing panel or the housing to form a rotating magnetic field; and a rotor rotated by the rotor, the one end of the input shaft being fixed to the rotor, wherein the drum body, the rear cover, and the front cover are formed of a same material, and a thickness of the drum body is set to be less than a thickness of the rear cover and a thickness of the front cover.
2. The apparatus of claim 1, wherein a thickness of the fixing panel is set to be greater than the thickness of the rear cover.
3. The apparatus of claim 2, further comprising: a ring-shaped panel bent portion formed by bending the fixing panel in a direction toward the rear cover or in a direction away from the rear cover to surround the output shaft.
4. The apparatus of claim 3, wherein the panel bent portion is formed by bending the fixing panel in the direction toward the fixing panel, wherein the housing is positioned inside the panel bent portion.
5. The apparatus of claim 4, further comprising: a drum bent portion formed by bending the rear cover in a direction toward the front cover or in a direction toward the fixing panel, the output shaft being fixed to the drum bent portion.
6. The apparatus of claim 1, wherein, among the thickness of the drum body, the thickness of the rear cover, and the thickness of the front cover, the thickness of the front cover is set to be the greatest.

7. The apparatus of claim 1, wherein, among the thickness of the drum body, the thickness of the rear cover, and the thickness of the front cover, the thickness of the rear cover is set to be the greatest.

8. The apparatus of claim 6, further comprising: a drum support rotatably supporting a circumferential surface of the front cover or a circumferential surface of the drum body.

9. The apparatus of claim 6, wherein the input shaft and the output shaft form a concentric axis, wherein the input shaft and the output shaft are parallel to a ground.

10. The apparatus of claim 6, wherein the input shaft and the output shaft form a concentric axis, wherein the input shaft and the output shaft are inclined at an angle of less than 90 degrees with respect to a ground.

11. The apparatus of claim 9, wherein the housing comprises: a housing base provided with an input shaft through-hole allowing the input shaft to be arranged therethrough in a penetrating manner; an input shaft support provided to the housing base to surround the input shaft through-hole; a housing circumferential surface extending from an edge of the housing base toward the fixing panel to define an accommodation space, the gear unit being mounted in the accommodation space; a cover body fixed to the housing circumferential surface to close the accommodation space; an output shaft through-hole formed through the cover body in a penetrated manner, the output shaft being inserted into the output shaft through-hole; an output shaft support protruding from the cover body toward the fixing panel to surround the output shaft through-hole; and an oil leakage preventer configured to prevent lubricating oil supplied to the gear unit from being discharged from the accommodation space.

12. The apparatus of claim 11, wherein the oil leakage preventer comprises: a first sealing portion disposed to seal a space between the input shaft support and the input shaft; a second sealing portion disposed to seal a space between the output shaft support and the output shaft; and a third sealing portion disposed to seal a space between the housing circumferential surface and the cover body.

13. The apparatus of claim 12, further comprising: a cover fixing plate provided on the housing circumferential surface, the cover fixing plate extending away from the housing circumferential surface in a radial direction of the input shaft through-hole such that the cover body is fixed to the cover fixing plate, wherein the third sealing portion comprises: a ring-shaped sealing protrusion protruding from the cover body toward the cover fixing plate; and a ring-shaped accommodation groove provided in the cover fixing plate, the sealing protrusion being accommodated in the accommodation groove.
